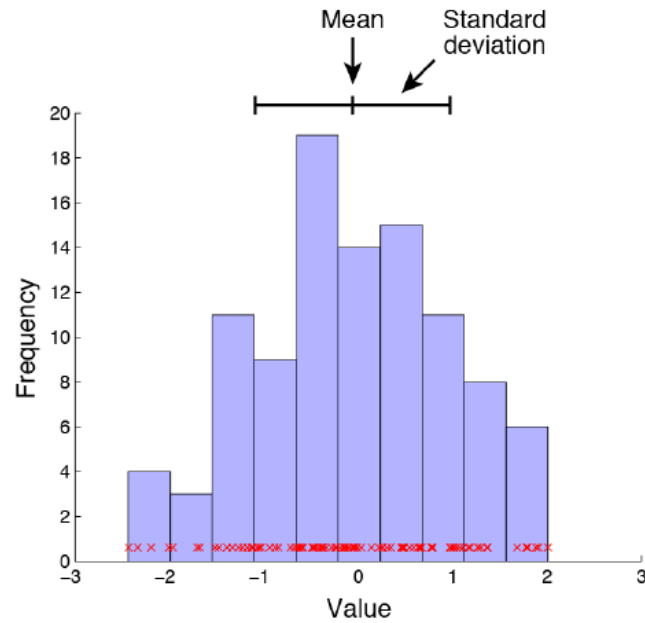


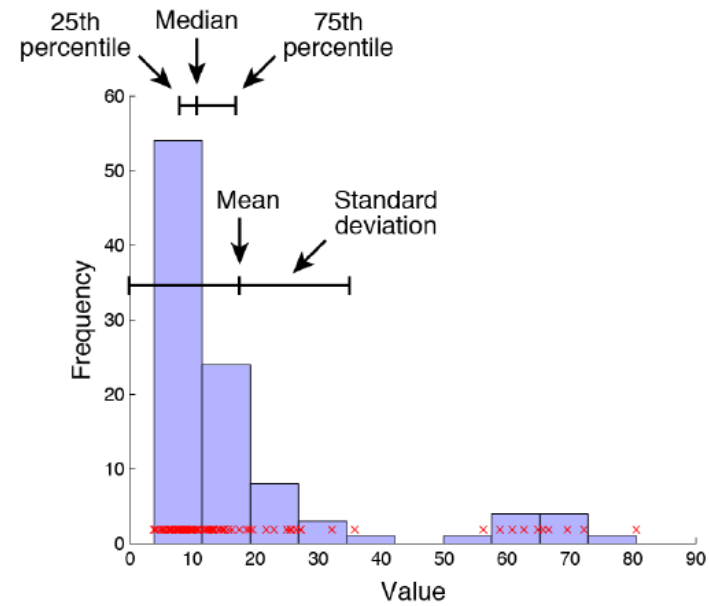
Descriptive statistics

Histogram of normal data



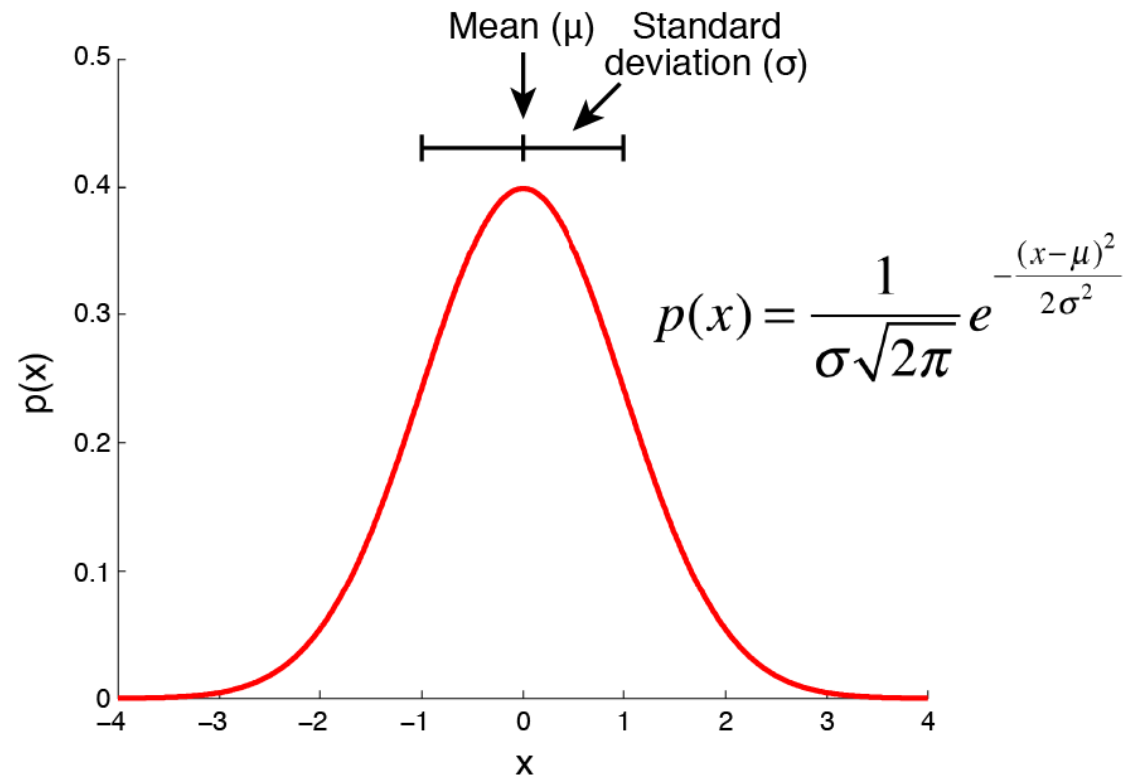
$$\text{mean}(x) = \bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad \text{std}(x) = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

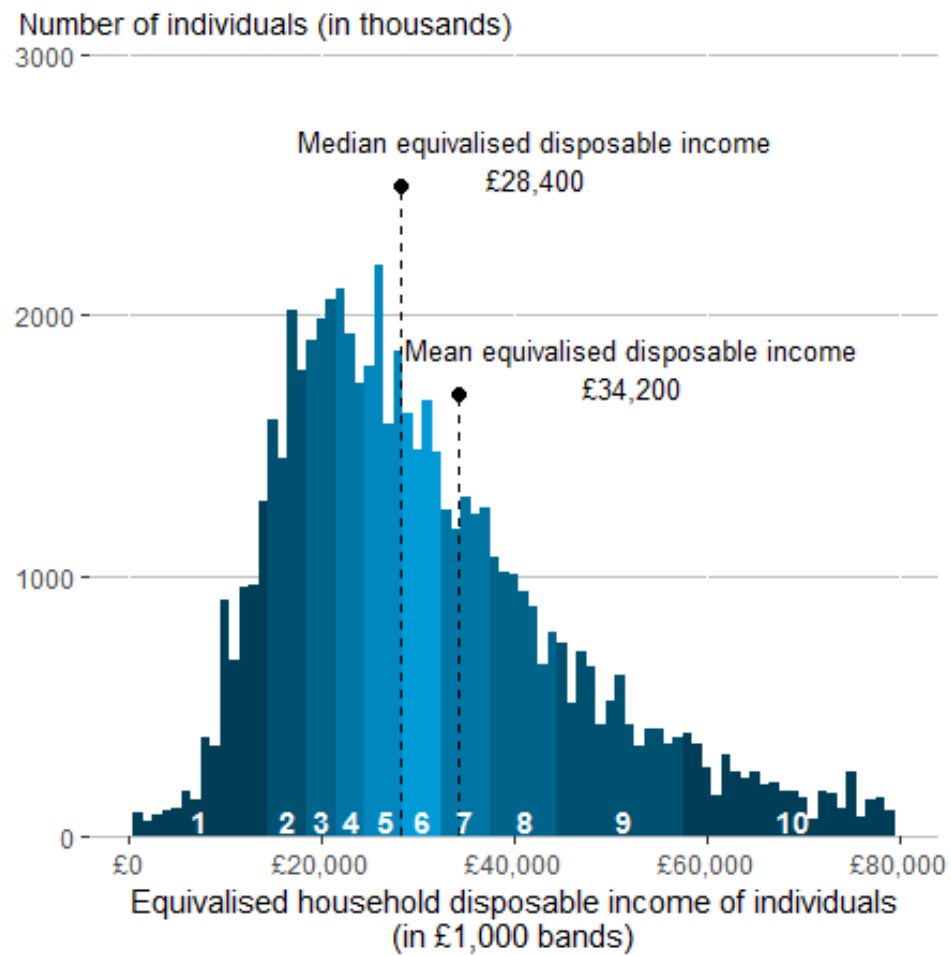
Histogram of non-normal data



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Gaussian probability distribution

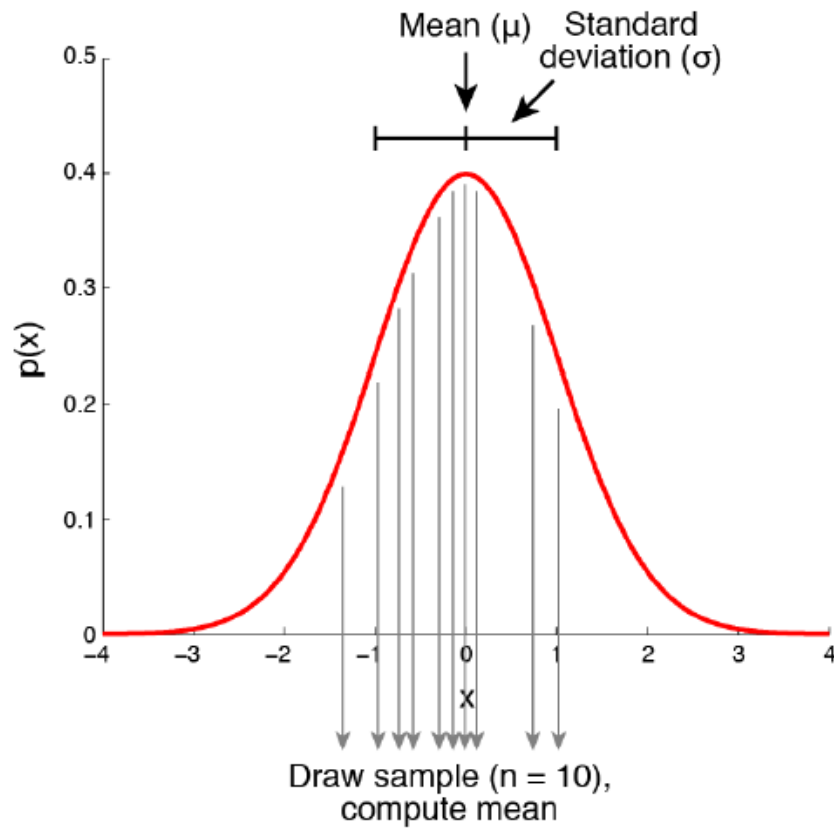




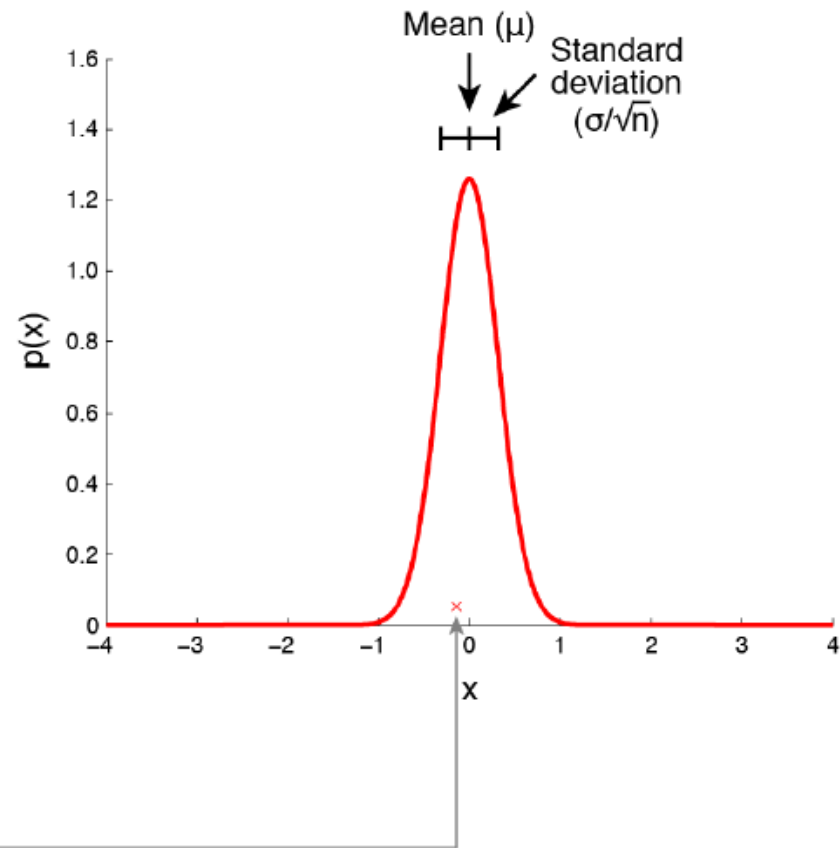
Data from 2018; Source: Office for National Statistics, UK

Hypothesis testing

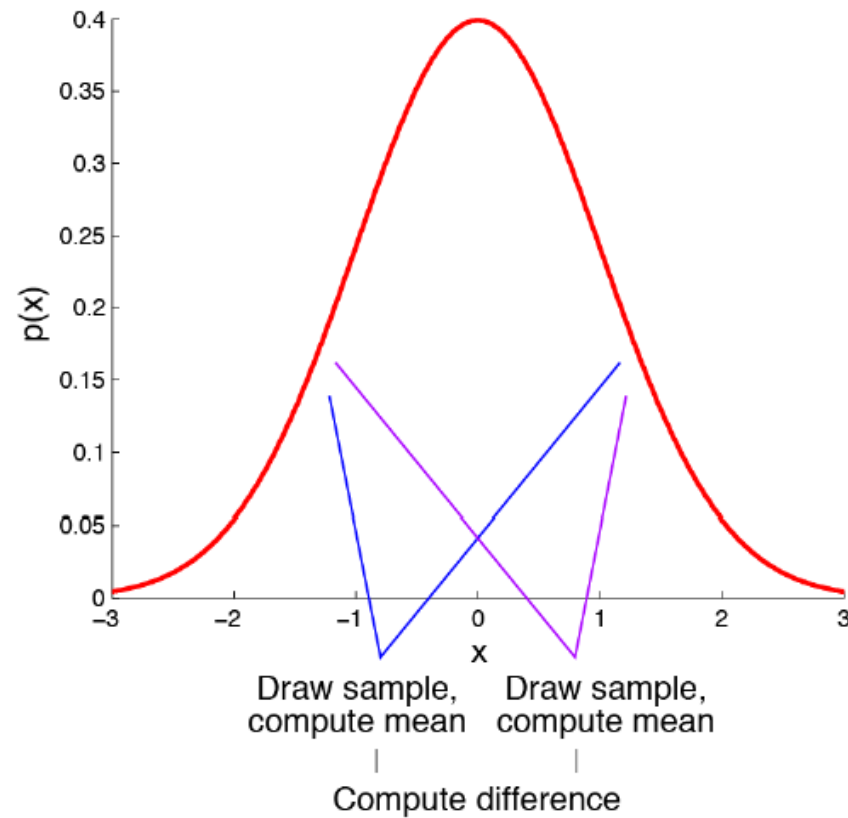
Population



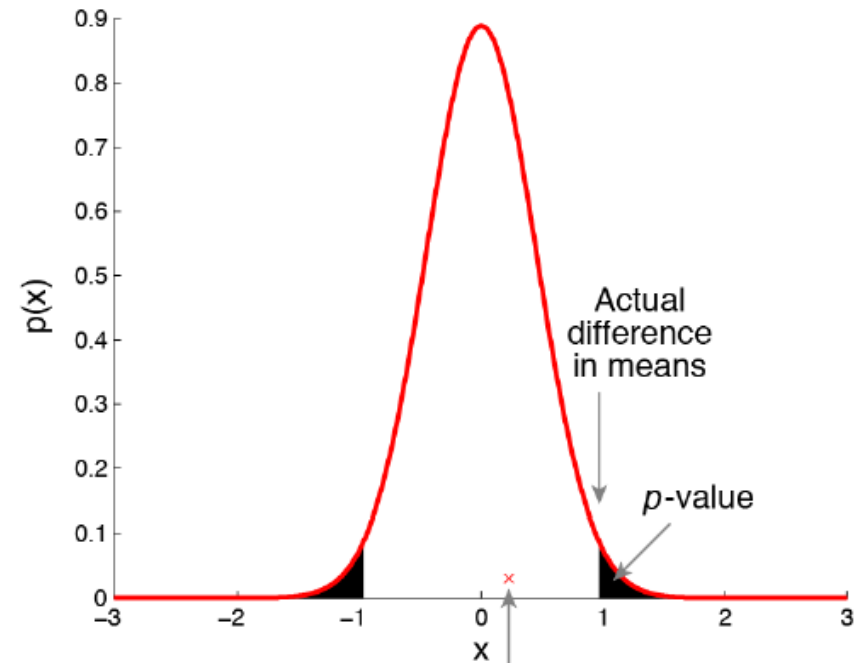
Sampling distribution of the mean

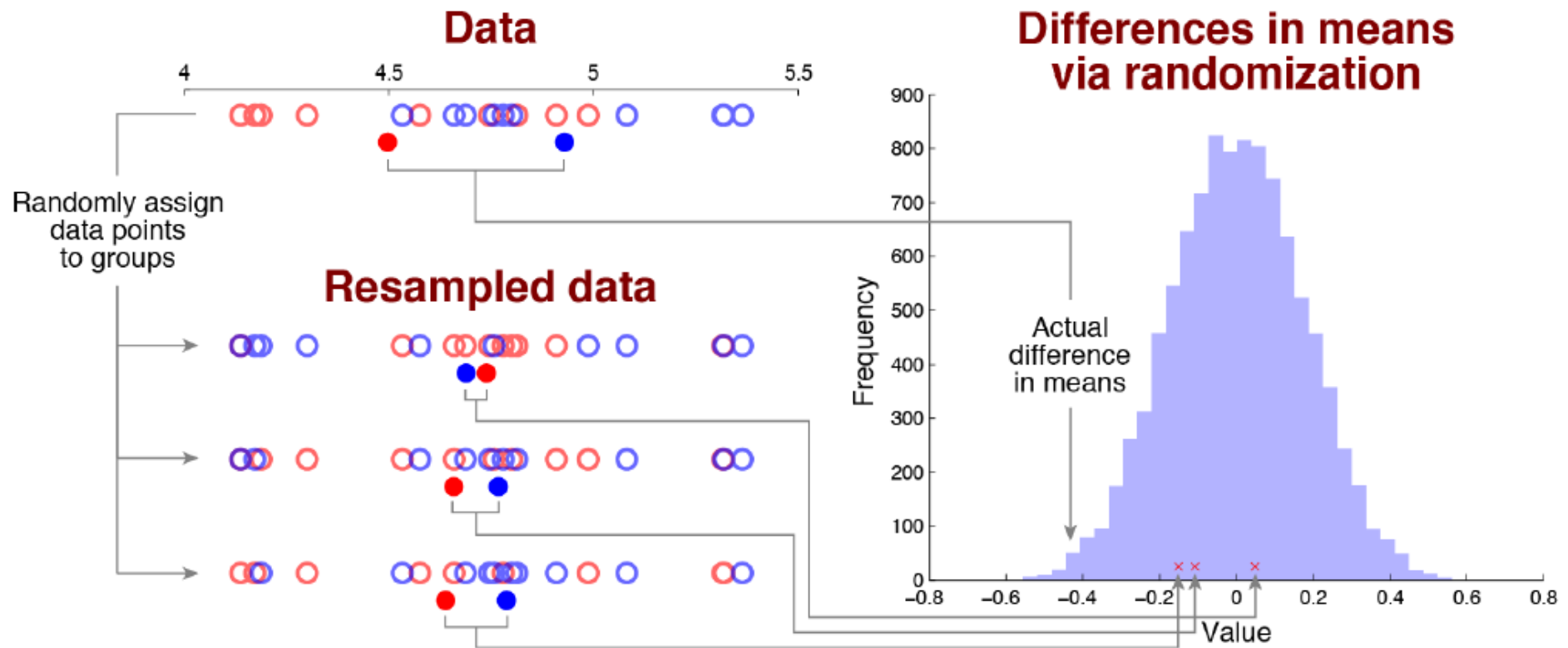


Distribution under null hypothesis



Sampling distribution of difference in means





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